

OMAK, WA, USA

WMO: 727890

Lat: 48.464N

Lon: 119.517W

Elev: 1300

StdP: 14.02

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 94197

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.5	11.0	-2.0	5.2	8.3	2.5	6.6	13.5	22.7	21.3	20.0	21.8	5.7	0	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.8	97.5	64.3	94.2	63.4	90.7	62.5	66.7	90.9	65.2	89.1	63.7	86.8	7.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.8	77.6	70.4	56.5	71.4	69.9	54.5	66.3	69.6	31.9	91.1	30.6	88.8	29.5	86.7	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	17.8	16.0	0.0	102.8	5.9	2.6	-4.2	104.7	-7.6	106.2	-10.9	107.6	-15.1	109.5		
(5)			-0.7	68.9	5.7	2.8	-4.8	70.9	-8.1	72.6	-11.3	74.2	-15.5	76.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	27.5	31.7	41.4	49.4	58.2	64.8	73.9	73.0	63.0	48.9	36.1	27.4		
	DBStd	17.73	8.09	6.89	6.64	6.35	6.97	6.63	6.48	6.12	6.99	7.03	7.90	7.69		
	HDD50	2808	698	513	274	88	13	1	0	0	2	101	418	700		
	HDD65	6277	1163	933	732	469	230	81	9	10	119	500	866	1165		
	CDD50	2702	0	0	7	69	266	445	742	712	391	68	2	0		
	CDD65	696	0	0	0	0	18	76	286	257	58	1	0	0		
	CDH74	10171	0	0	0	30	437	1252	4075	3468	887	22	0	0		
	CDH80	4926	0	0	0	3	127	502	2191	1772	328	3	0	0		
	WSAvg	6.7	4.8	5.9	7.0	7.6	7.4	7.5	8.0	7.8	7.7	6.6	5.6	4.9		
	PrecAvg	11.4	1.1	1.2	0.9	0.9	1.0	0.9	0.4	0.7	0.6	0.7	1.4	1.9		
(15) Precipitation	PrecMax	15.3	2.5	3.0	1.8	2.4	2.8	3.2	1.4	2.1	1.5	2.4	3.6	4.6		
	PrecMin	8.3	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3		
	PrecStd	1.9	0.7	0.8	0.6	0.7	0.7	0.7	0.4	0.7	0.6	0.7	0.9	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.9	53.0	66.1	78.6	88.5	97.0	102.1	100.5	93.6	77.1	57.5	46.3		
		MCWB	39.4	42.6	50.3	56.8	62.2	64.7	65.7	65.5	62.6	57.9	49.2	41.0		
	2%	DB	41.3	48.0	61.6	72.5	83.6	90.8	98.8	97.1	88.4	70.0	53.0	42.6		
		MCWB	36.9	40.9	47.2	53.5	60.1	62.8	65.0	64.0	61.2	53.8	46.3	38.8		
	5%	DB	37.6	44.7	57.2	68.3	79.0	85.6	95.4	93.2	83.7	65.1	49.9	38.8		
		MCWB	35.3	38.5	45.1	51.3	57.9	60.6	63.7	63.1	59.6	51.6	44.9	36.0		
	10%	DB	36.0	41.1	54.0	64.0	74.2	81.3	91.5	90.1	79.1	61.7	47.2	36.3		
		MCWB	34.5	36.8	43.5	48.8	55.8	59.4	62.9	62.3	57.6	49.7	43.0	34.3		
	0.4%	WB	40.2	45.4	51.9	58.0	64.2	67.1	69.5	68.3	65.4	59.3	50.7	42.4		
		MCDB	44.9	50.2	64.0	76.6	83.8	90.2	93.7	94.3	87.2	71.5	55.5	45.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.6	41.8	48.6	54.5	61.5	64.4	67.3	66.2	62.6	55.5	48.0	39.3		
		MCDB	40.0	46.2	59.4	70.0	79.5	86.6	91.5	90.9	84.2	64.8	51.3	42.2		
	5%	WB	36.1	39.2	46.4	52.1	59.3	62.5	65.6	64.5	60.5	53.2	46.0	36.5		
		MCDB	37.7	43.4	55.6	66.3	75.4	82.4	90.2	88.6	80.6	62.9	48.7	38.2		
	10%	WB	34.5	37.2	44.4	49.8	57.2	60.4	64.0	63.1	58.6	50.9	43.7	34.5		
		MCDB	35.6	40.5	52.5	62.1	72.0	78.0	87.7	86.4	76.7	59.8	46.7	35.8		
	5% DB	MDBR	9.2	14.3	20.7	25.1	26.6	27.0	30.8	30.6	28.7	22.3	13.9	9.4		
		MCDBR	11.5	18.6	26.2	31.1	33.7	33.7	35.4	34.9	34.7	26.8	17.3	12.8		
		MCWBR	8.9	12.4	15.4	16.0	15.2	13.0	12.5	12.7	14.8	14.3	11.7	9.9		
		MCWBR	10.3	16.3	24.4	29.2	30.4	30.9	32.7	31.9	31.8	24.0	14.4	10.9		
(40) Clear-Sky Solar Irradiance	taub		0.282	0.285	0.304	0.326	0.347	0.348	0.341	0.344	0.320	0.314	0.296	0.280		
		taud	2.354	2.407	2.452	2.450	2.429	2.459	2.490	2.473	2.539	2.538	2.508	2.398		
	Ebn at Noon		250	279	291	294	290	289	290	285	283	266	247	236		
		Edh at Noon	22	26	30	33	35	34	33	32	27	23	19	18		
(44) All-Sky Solar Radiation	RadAvg		339	668	1041	1514	1852	2001	2286	1915	1403	829	398	268		
	RadStd		42	74	84	110	135	155	134	124	102	91	52	34		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	N/A
(47) Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	+61

Nomenclature: See separate page